



# Entrepreneurship and sustainable development

# Entrepreneurship?

- The transformation of human intellectual labor into economic value.





# Concepts of Entrepreneurship



- **Enterprise:** An economic unit that meets the demands and needs of people by producing and marketing products & services for specific purposes.
- **Management:** Planning, organizing, conduct and controlling of the resources.
- **SME (Small and Medium Enterprise):** A company, or companies considered as a group, that are neither very small nor very large.

# Scope of an Enterprise



**VALUE**



## PARTNERS

- Customer
- Financier
- Worker
- Supplier
- Distributer
- Government
- Nature

## EXPECTATIONS

- ➔ High value
- ➔ High profit
- ➔ Satisfying job conditions
- ➔ Profitable relationships
- ➔ Profitable relationships
- ➔ Obeying tax rules
- ➔ Passing on posterity

## INNER OBJECTIVES





# Management Functions



- **Planning:** In accordance with the future goals of the business, the manager reveals how these goals can be achieved.
- **Organizing:** The Manager determines the organizational structure and what works and by whom in this structure, in order to carry out dozens of different activities within the enterprise according to a certain plan.
- **Conduct:** The manager keeps the motivation of the employees high, leads them effectively and designs the communication channels well in order to achieve the unity of purpose among the employees.
- **Controlling:** The manager checks whether the initially targeted results have been achieved. If there is a deviation, the reason is determined and transferred to a new planning process as feedback.



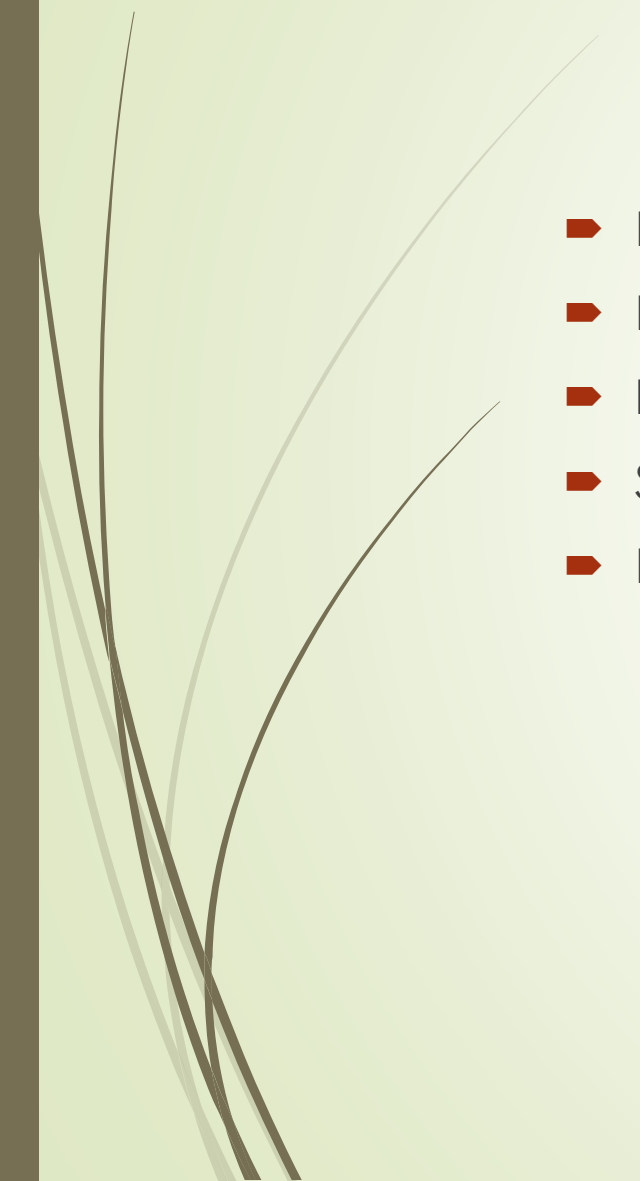


# Advantages of SMES

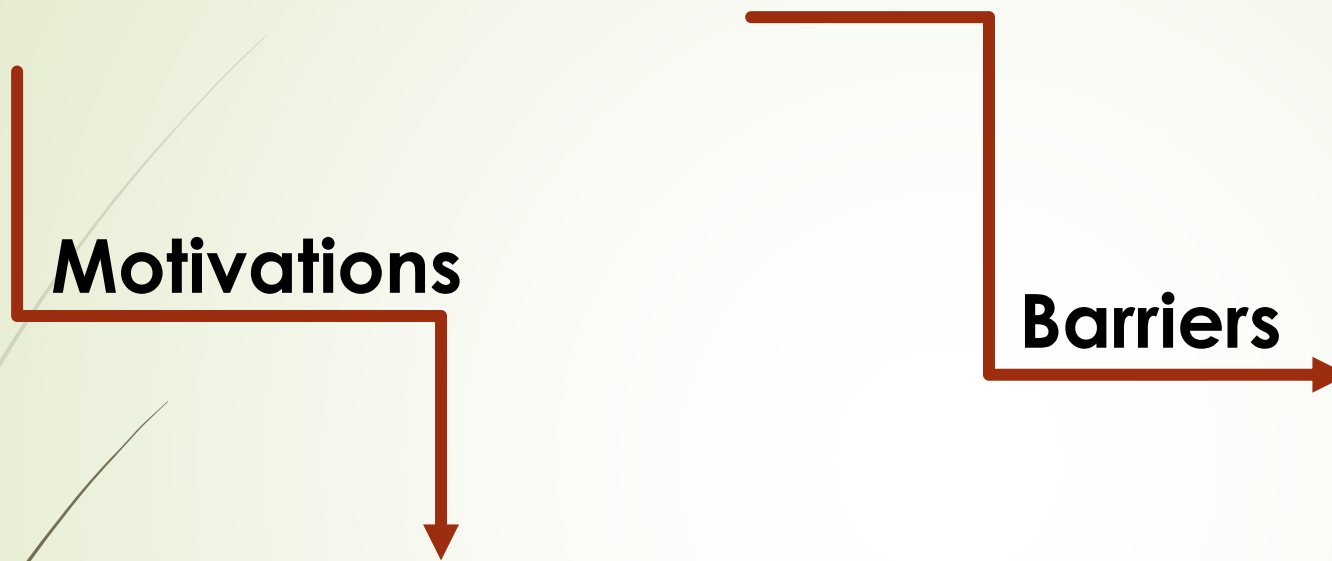
- Supply, production and marketing systems are more flexible according to consumer preferences.
- They can operate in low-potential markets.
- They are closer to the market and customers.
- They can better identify and evaluate innovation opportunities.
- They are more efficient in technological innovations.
- They can increase the motivation of the employees by establishing close and sincere relationships.
- They can quickly fill the gaps left by large enterprises in the market.
- There is less bureaucracy.
- They can be established with small savings and show the entrepreneurial potential of the society.
- They can benefit more from subsidized funds.
- They are complementary to large businesses.



# Disadvantages of SMES

- Lack of knowledgeable and experienced staff
  - High production costs due to low quantity supply
  - Insensitivity to price competition of large enterprises
  - Sales fluctuations are high
  - Dependence on a single manager in decision making
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# Entrepreneurship



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graph TD; A[Entrepreneurship] --> B[Motivations]; A --> C[Barriers]; B --> D["To secure the future of the person herself and her family<br/>To have a good standard of living and make more money<br/>To prove that a person can do something, to ensure personal development, to be known in the public, a sense of accomplishing a difficult job<br/>To be able to make one's own decisions, protect personal freedom, be the boss of her own business and offer job security"]; C --> E["Long working hours<br/>Irregular Income<br/>A Stressful Life"];
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## Motivations

- To secure the future of the person herself and her family
- To have a good standard of living and make more money
- To prove that a person can do something, to ensure personal development, to be known in the public, a sense of accomplishing a difficult job
- To be able to make one's own decisions, protect personal freedom, be the boss of her own business and offer job security

## Barriers

- Long working hours
- Irregular Income
- A Stressful Life



# Entrepreneurship Progress

## Identification of Opportunities

- Changes in economic factors
- Technological changes
- Changes in demographic trends
- Changes in the legal environment

## Business Model and Business Plan Development

- Business Model: A brief description of how the business will create value, how to deliver this value to the customer and generate revenue.
- Business Plan: Business-related technical, economic and financial feasibility studies are covered in a comprehensive way.

## Finding Resources

- Human
- Fund
  - Equity capital
  - Friends and family
  - Angel investors
  - Bank credits
  - Public funds
- Means of Production

## Growth and Exit

- Continuing the business
- Handing on the business



# Characteristics of a Successful Entrepreneur

- **Self-sufficiency:** The belief of a person in the knowledge and abilities that he/she can exhibit in achieving a desired result or in a particular situation.
- **Innovativeness:** The innovation includes not only products but also services, creating new markets, entering new distribution channels, and differentiating processes.
- **Risk Taking:** This does not mean taking excessive or high risks like a gambler. Risk taking is a matter of decision making. The entrepreneur should be able to minimize the risk by reading the changes in her environment correctly and by renewing herself.
- **Leadership:** From the business point of view, leadership is the characteristics that can direct the behaviors of the employees in line with the business objectives and persuade them to show the behaviors that will achieve these goals.

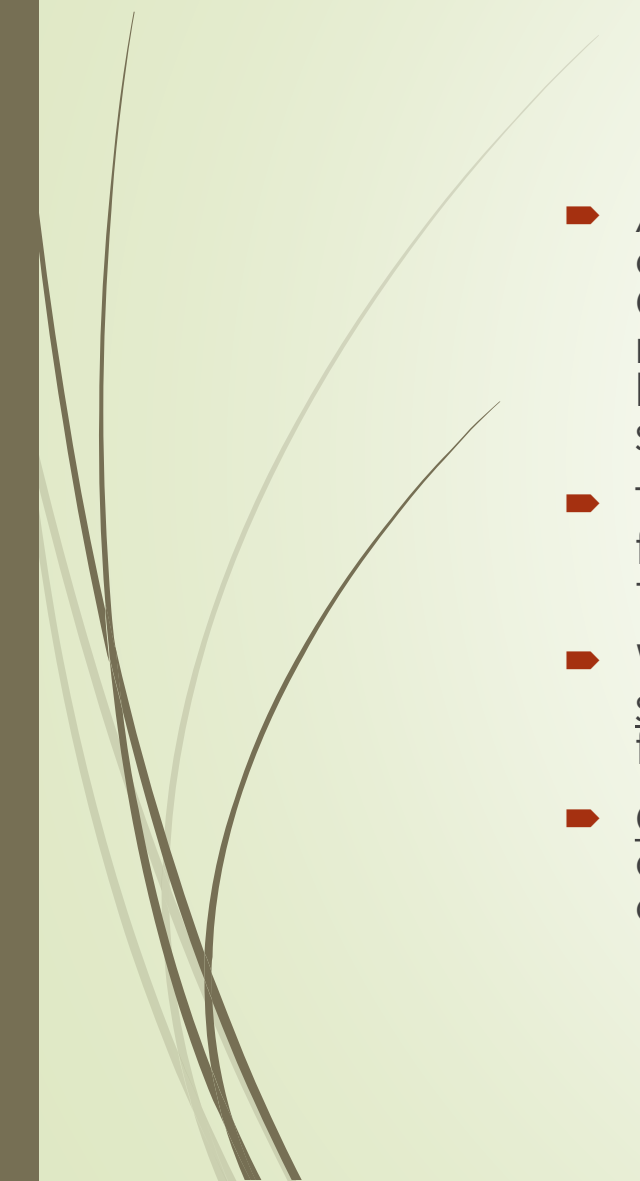
# Innovation



- Innovation means advantageous
- Speaking in the context of entrepreneurship, it can be defined as innovations that will make a difference in the market and provide competitive advantage.
- Innovation can also be made in areas such as business model, organizational, business network, product performance, sales channel, customer experience, and product presentation.



# Innovation Areas



- As an example of innovation in terms of business model, offering basic features called "semi-paid" free of charge and charging extra features can be shown. Open source can be shown as an example of innovation in terms of business model. Developing products as open source, including the customer and business network in the business model, and charging for consultancy and special solutions is innovation in terms of business model.
- The customer experience offered by Starbucks in the coffee market and the fact that it offers a different service to its customers by likening its branches to the home environment is an innovation in terms of customer experience.
- With the rapid growth of the Internet, we come across innovation in terms of sales channels every day. It may be an example to buy products we bought from grocery stores, markets and stores on the Internet.
- Organizational innovation is an example of outsourcing, more use of digital tools and providing advantages that will stand out in the market with organizational changes.



# Innovation Resources



- ▶ **Customer:** Listening to the customer and focusing on the customer's needs and problems, not on the customer's solutions, is prominent for innovation.
- ▶ **Technology:** An example of technology-based innovation can be explained with Nanotechnology. Technological developments in nanotechnology enable products and innovations using nanotechnology to be produced.
- ▶ **Market trends:** Innovation based on market trends, on the other hand, is the innovation created in the light of market developments. Making local versions of global projects (not verbatim copies) can be cited as a prime example for this type of innovation.





# Sustainable Development

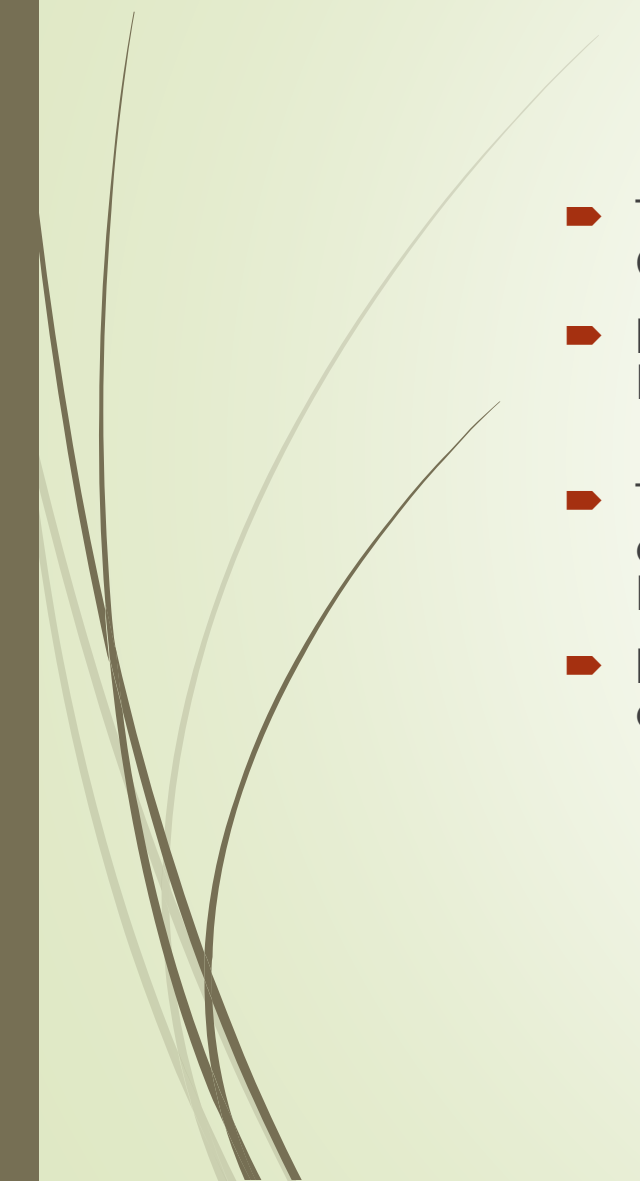


- Sustainable development is the organizing principle for meeting human development goals while simultaneously sustaining the ability of natural systems to provide the natural resources and ecosystem services on which the economy and society depend.
- To put it another way, United Nations descriptions over sustainable development is understood as satisfying the needs of the present generation without compromising the needs of future generations.
- The desired result is a state of society where living conditions and resources are used to continue to meet human needs without undermining the integrity and stability of the natural system.





# Sustainable Development



- The term “sustainable development” first came to prominence in the World Conservation Strategy (WCS) in 1980.
- It achieved a new status with the publication of two significant reports by Brundtland Commission on: “North and South : a programmed for survival (1980)” and “Common Crisis (1983)
- Term sustainable development was popularized by the World Commission on Environment and Development in its 1987 report entitled “Our Common Future”
- It has gained even greater attention since the United Nations Conference on Environment and Development (UNCED) held in Rio de Janeiro in June 1992.



# Major Components

- Major components of Sustainable Development:
  - Environmental (Ecological)
  - Economic
  - Politics
  - Culture (Social)



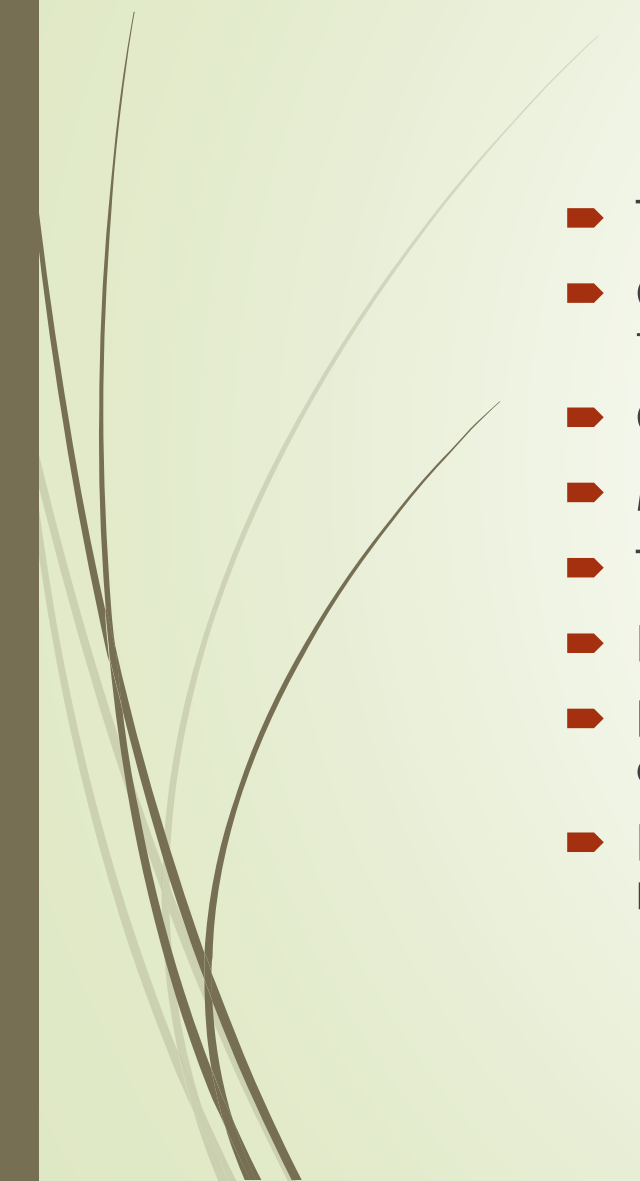
# Environmental Sustainability



- The ecological stability of human settlements is part of the relationship between humans and their natural, social and built environments.
- Fundamental human needs such as the availability and quality of air, water, food and shelter are the ecological foundations for sustainable development.
- Addressing public health risk through investments in ecosystem services can be a powerful and transformative force for sustainable development which, in this sense, extends to all species.

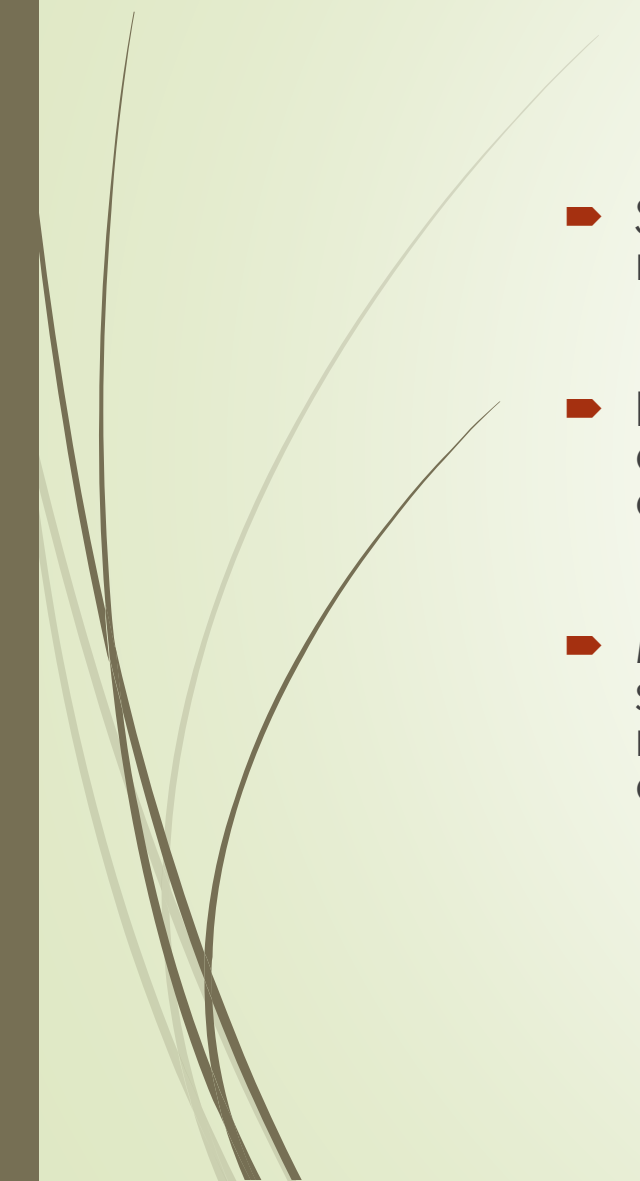


# Achieving Environmental Sustainability

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- The 3Rs of Sustainable Waste Management : Reduce, Reuse, Recycle
  - Cradle-to-Cradle Manufacturing : Creating products that are designed for upcycling
  - Control of carbon
  - Maintaining “stocks” of natural resources
  - Transitioning to renewable energy sources
  - Responsible consumption
  - Developing new technologies (green technologies, renewable energy, etc.)
  - Reorganizing living conditions in the form of eco-villages, eco-municipalities, and sustainable cities



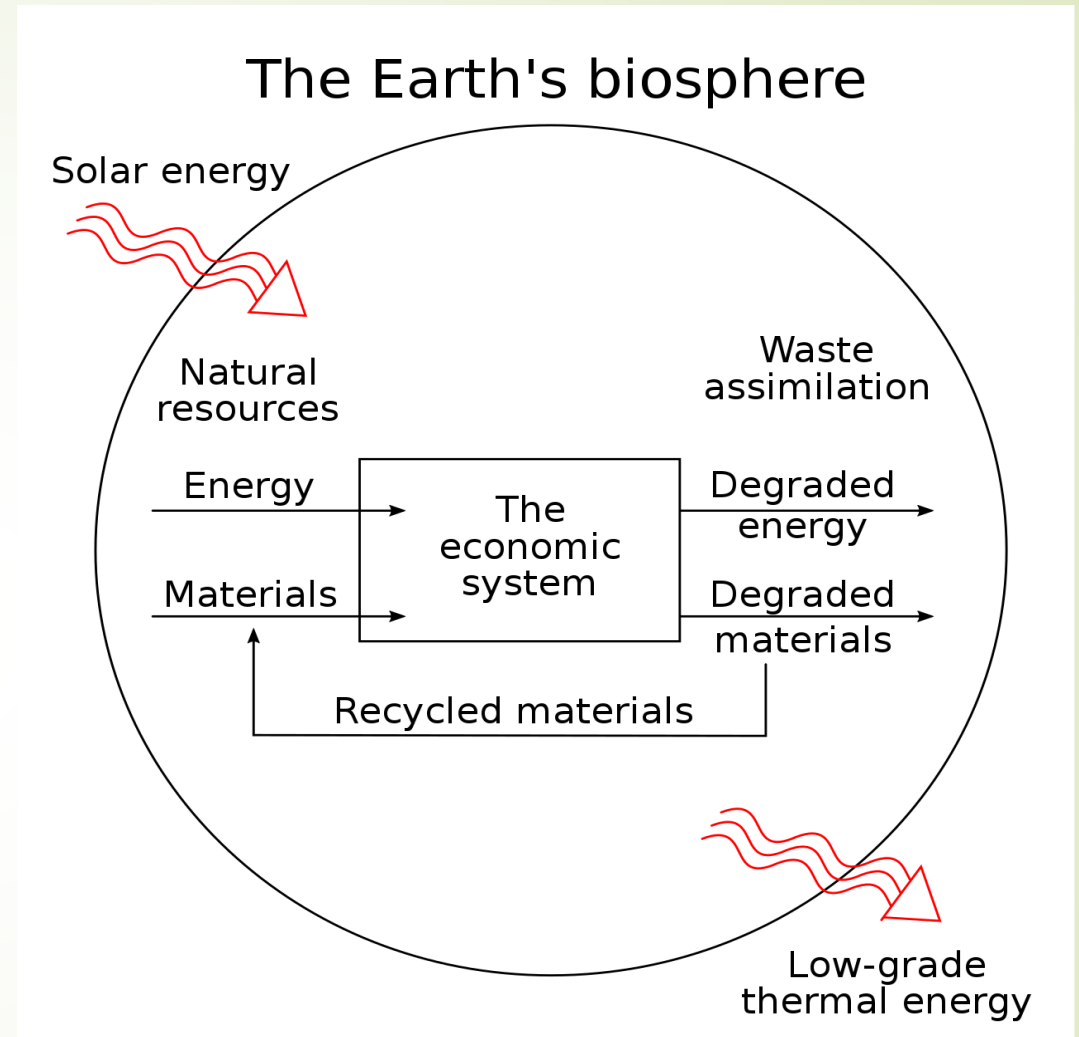
# Economics



- Sustainable development involve improvements in the quality of life for many but may necessitate a decrease in resource consumption
- It has been suggested that because of rural poverty and overexploitation, environmental resources should be treated as important economic assets, called natural capital
- Main goals of economic systems are maximizing the amount of goods and services produced in the techno sphere, maximizing the efficient use of mineral resource flows, biological, energy and information flows; adaptation technologies mineral resources and reprocessing products.



- Humanity's economic system viewed as a subsystem of the global environment
- Financial objectives of sustainable development are :
  - Reduced utility bills & operating costs by specifying energy, water & resource efficient products, services & buildings
  - Lower waste management fees through the procurement of environmentally preferable products





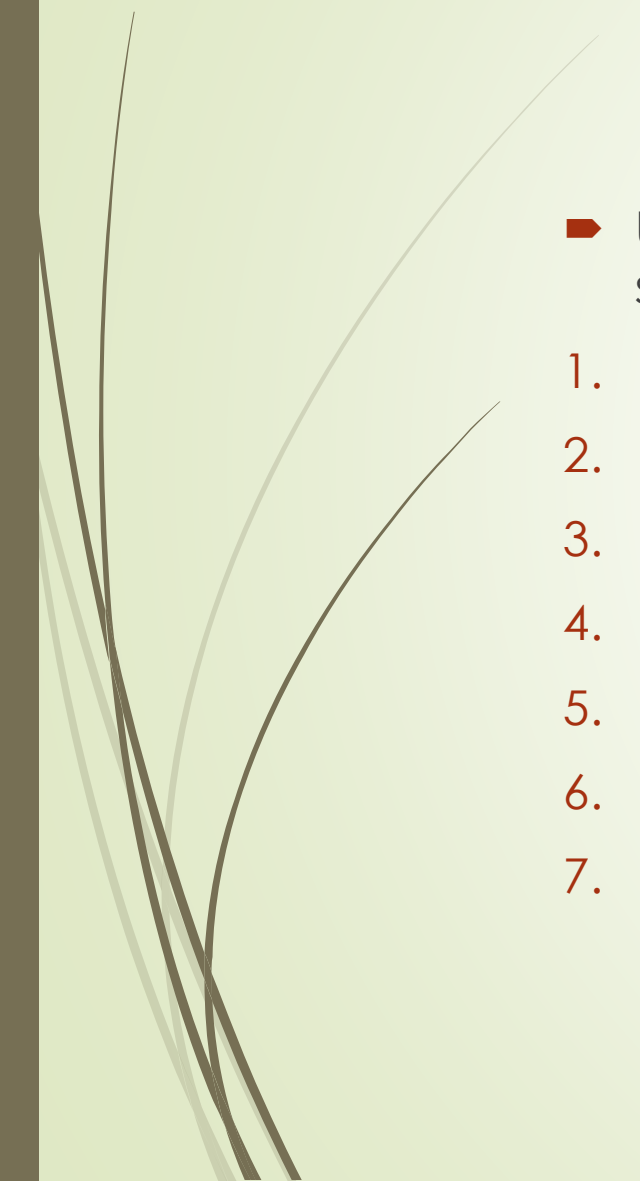
# Politics



- ▶ The United Nations Global Compact Cities Programme has defined sustainable political development in a way that broadens the usual definition beyond states and governance.
- ▶ The political is defined as the domain of practices and meanings associated with basic issues of social power as they pertain to the organization, authorization, legitimation and regulation of a social life held in common.
- ▶ This definition is in accord with the view that political change is important for responding to economic, ecological and cultural challenges. It also means that the politics of economic change can be addressed.



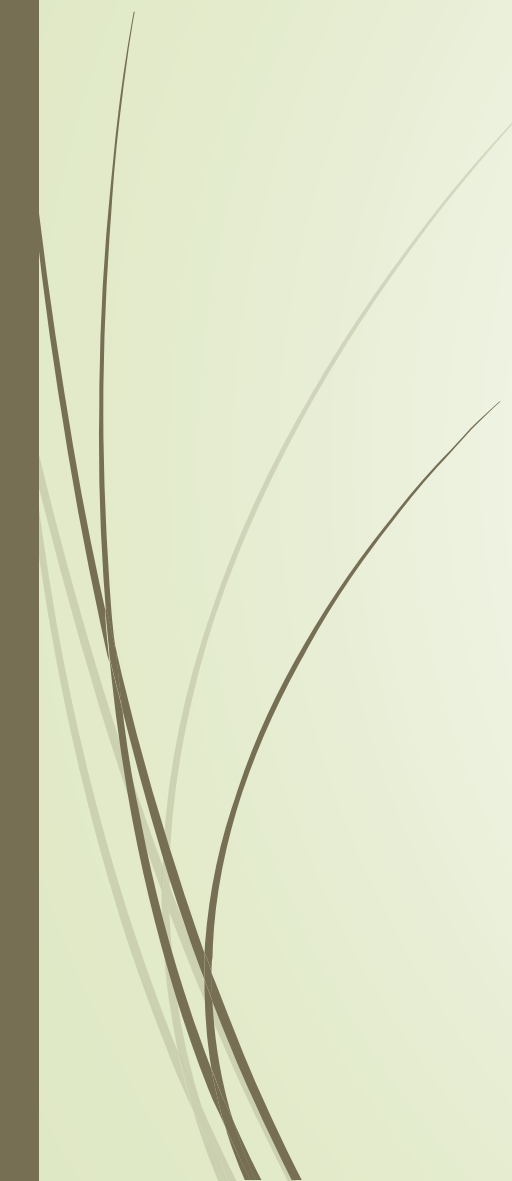
# Politics



- United Nations Global Compact Cities Programme have listed seven subdomains of the domain of politics
- 1. Organization and governance
- 2. Law and justice
- 3. Communication and critique
- 4. Representation and negotiation
- 5. Security and accord
- 6. Dialogue and reconciliation
- 7. Ethics and accountability



# Culture



- ▶ Culture have pointed out that a fourth dimension should be added to the dimensions of sustainable development.
- ▶ Cultural regulations inaugurates a new perspective and points to the relation between culture and sustainable development;
- ▶ Different societies have different values systems and cultural traditions. The concept such as justice, equity, non-violence, etc. are culture bound.



# The Global Goals

- In 2015, world leaders (United Nations) agreed to 17 Global Goals (officially known as the Sustainable Development Goals or SDGs).
- These goals have the power to create a better world by 2030, by ending poverty, fighting inequality and addressing the urgency of climate change.



# Sustainable Development Goals



- 17 goals
- 169 targets

<https://sustainabledevelopment.un.org/sdgs>





# Traditional Engineering vs. Sustainable Engineering

| Traditional Engineering                                             | Sustainable                                                                                              |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Considers the object                                                | Consider the system in which object will be used                                                         |
| Focuses on technical issues                                         | Integrates technical and non-technical issues                                                            |
| Solves the immediate problem (now)                                  | Strives to solve the problem for the indefinite future (forever)                                         |
| Considers the local context (users)                                 | Consider the global context (planet)                                                                     |
| Assumes others will deal with politics, ethics, and societal issues | Acknowledges the need for engineers to interact with experts in other disciplines to the related problem |



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